CHEM 1301 Driller 9-1 — Answer Key

Worked solutions with course-material citations

Every numeric answer was recomputed by `_scripts/build_driller_9_1.py`, which builds this key and the driller from the same data. Atomic masses are periodic-table values (H 1.008, C 12.01, N 14.01, O 16.00, …). Citations name the slide (same number in the `.pdf` and `.pptx`) where the content is taught, plus the closest homework screenshot.

How to read the solutions. Each calculation is a chain of conversion factors written want-over-have. The label under each factor says where its numbers come from (a molar mass, or two coefficients from the balanced equation), and the crossed-out units are the ones that cancel. Study guide section 9.0 explains this format.

Score bands. 9–10 solid on Chapter 9 · 7–8 rework the study guide sections for the questions you missed · 6 or below go back through study guide sections 9.2–9.7 before the next driller.

Q	Ans	Answer	Type	Topic (source)
1	**a**	2 mol of H_2 reacts with 1 mol of O_2 to form 2 mol of H_2O	concept	What the coefficients mean (Ch 9 sl. 2–5)
2	**b**	$\frac{2 \text{ mol } Fe_2O_3}{3 \text{ mol } O_2}$	concept	Choosing the mole ratio (Ch 9 sl. 5–8)
3	**e**	12.0 mol	calc	Mole–mole conversion (Ch 9 sl. 10–11)
4	**d**	grams A → moles A → moles B → grams B	concept	Order of the stoichiometry steps (Ch 9 sl. 13–14)
5	**c**	23.5 g	calc	Grams to grams (Ch 9 sl. 14, 17–19)
6	**c**	0.998 mol	calc	Grams to moles of another substance (Ch 9 sl. 18)
7	**e**	16.0 g	calc	Balance first, then grams of a reactant (Ch 9 sl. 9, 16, 19)
8	**a**	CO is limiting, and 0.5 mol of O_2 is left over	calc	Limiting reactant and leftover (Ch 9 sl. 21–23)
9	**b**	43.9 g	calc	Theoretical yield from two masses (Ch 9 sl. 24–27)
10	**d**	88.5%	calc	Percent yield (Ch 9 sl. 28–32)

Mix: 3 concept / 7 calculation · Ch 9: 10

1. What the coefficients mean — a) 2 mol of H_2 reacts with 1 mol of O_2 to form 2 mol of H_2O

Source: Chapter 9 post, slide 2 (coefficients give relative numbers of molecules), slide 3 (Table 9.1: molecules, dozens, moles), slide 5
Homework match: HW22 — hw22ex1.png ($H_2 + F_2 \rightarrow 2HF$ read in molecules and in moles)
What it tests: Reading a balanced equation in moles: coefficients are mole (particle) ratios, not masses.

Step 1: read the coefficients. H_2 **2**, O_2 **1** (nothing written means 1), H_2O **2**.

Step 2: say it in moles. Slide 3 (Table 9.1) shows that the same coefficients can be read as molecules, dozens, or moles. In moles: **2 mol H_2 + 1 mol O_2 → 2 mol H_2O**.

Coefficients never mean grams. 2 mol H_2 (4.03 g) and 1 mol O_2 (32.00 g) have very different masses.

Why the other choices are wrong:

- b) 2 molecules of H_2 react to form 1 molecule of H_2O: H_2O's coefficient is 2, so 2 molecules of H_2 form **2** molecules of H_2O.
- c) 2 g of H_2 reacts with 1 g of O_2 to form 2 g of H_2O: Coefficients count molecules or moles, never grams (slide 2).
- d) The mass of H_2O formed equals the mass of H_2 used: Mass is conserved for the whole reaction: mass of H_2 + mass of O_2 = mass of H_2O.
- e) 1 mol of H_2 reacts with 2 mol of O_2: The ratio is flipped. H_2's coefficient is 2 and O_2's is 1 (no number written means 1).

2. Choosing the mole ratio — b) $\dfrac{2\ \textbf{mol } Fe_2O_3}{3\ \textbf{mol } O_2}$

Source: Chapter 9 post, slides 5–8 (mole ratios from the coefficients: $N_2 + 3H_2 \rightarrow 2NH_3$, $N_2 + O_2 \rightarrow 2NO$)
Homework match: HW22 — hw22ex2.png (which mole ratio for $4B + 3O_2 \rightarrow 2B_2O_3$: 3 mol O_2 / 4 mol B)
What it tests: Building a mole ratio from two coefficients and orienting it want-over-have.

Step 1: pick the two substances. You **have** O_2 and **want** Fe_2O_3. Fe is not involved.

Step 2: read their coefficients. O_2 **3**, Fe_2O_3 **2**. So 3 mol O_2 makes 2 mol Fe_2O_3.

Step 3: orient it want-over-have. Fe_2O_3 (want) on top, O_2 (have) on the bottom:

$$\text{mol } O_2 \times \frac{2 \text{ mol } Fe_2O_3}{3 \text{ mol } O_2} \text{ leaves mol } Fe_2O_3$$

Why the other choices are wrong:

- a) $\dfrac{3 \text{ mol } O_2}{2 \text{ mol } Fe_2O_3}$: Upside down. The unit you have (mol O_2) has to be on the bottom so it cancels.
- c) $\dfrac{2 \text{ mol } Fe_2O_3}{4 \text{ mol Fe}}$: That converts mol Fe to mol Fe_2O_3. The question starts from O_2.
- d) $\dfrac{3 \text{ mol } O_2}{4 \text{ mol Fe}}$: That relates the two reactants (O_2 and Fe). It never reaches Fe_2O_3.

- e) $\dfrac{32.00\text{ g }O_2}{1\text{ mol }O_2}$: That is the molar mass of O_2. It converts moles to grams of the **same** substance, not to another substance.

3. Mole–mole conversion — e) 12.0 mol

Source: Chapter 9 post, slide 11 (Mole–Mole Conversions Practice 2: $2C_2H_6 + 7O_2$), slide 10
Homework match: HW22 — hw22ex3.png (moles of one substance from moles of another, e.g. mol Se from mol H_2O)
What it tests: Picking the right two coefficients out of a four-substance equation.

Step 1: coefficients you need. Have C_3H_8 (coefficient **1**, nothing written); want O_2 (coefficient **5**). CO_2 and H_2O are not involved.

Step 2: multiply by the mole ratio (want over have).

$$2.40\ \cancel{\text{mol } C_3H_8} \times \underbrace{\frac{5\text{ mol }O_2}{1\ \cancel{\text{mol } C_3H_8}}}_{\text{coefficient of } O_2\ /\ \text{coefficient of } C_3H_8} = \boxed{12.0\text{ mol }O_2}$$

Why the other choices are wrong:

- a) 0.480 mol: Ratio upside down: $2.40 \times 1/5$.
- b) 2.40 mol: Assumed 1 : 1. O_2's coefficient is 5.
- c) 7.20 mol: Used CO_2's coefficient (3) instead of O_2's (5).
- d) 9.60 mol: Used H_2O's coefficient (4) instead of O_2's (5).

4. Order of the stoichiometry steps — d) grams A → moles A → moles B → grams B

Source: Chapter 9 post, slide 14 (Steps for Calculating the Masses of Reactants and Products), slide 13, slide 16
Homework match: HW23 — hw23ex1.png (mass of C_2H_2 from 3.25 g CaC_2)
What it tests: The four-step path of slide 14: moles are always the middle step, and the coefficients act only on moles.

Slide 14's steps after balancing: convert grams to moles, use the mole ratio, then convert moles to grams.

$$\text{g A} \xrightarrow{\times \frac{1\text{ mol A}}{\text{molar mass of A}}} \text{mol A} \xrightarrow{\times \frac{\text{coef. B}}{\text{coef. A}}} \text{mol B} \xrightarrow{\times \frac{\text{molar mass of B}}{1\text{ mol B}}} \text{g B}$$

The mole ratio is the **only** link between A and B, and it only works on moles. So there has to be a moles step on each side of it.

Why the other choices are wrong:

- a) grams A → moles B → moles A → grams B: You can't get moles of B directly from grams of A. Convert to moles of A first.
- b) grams A → grams B, using the coefficients as a mass ratio: Coefficients are mole ratios, never mass ratios (slide 2).

- c) grams A → moles A → grams B: Skips the mole ratio. Moles of A are not moles of B unless the coefficients are equal.
- e) grams A → moles A → moles B → atoms B: Ends in particles. The question asks for grams, so the last step multiplies by B's molar mass.

5. Grams to grams — c) 23.5 g

Source: Chapter 9 post, slides 17–19 (Stoichiometry Practice: g → mol → mol → g), slide 14 (steps)
Homework match: HW23 — hw23ex2.png (grams of Br_2 from the decomposition of 34.0 g BrF_3)
What it tests: The full g → mol → mol → g chain with a 2 : 1 ratio.

Step 1: molar masses. $H_2O_2 = 2(1.008) + 2(16.00) = 34.02$ g/mol. $O_2 = 2(16.00) = 32.00$ g/mol (O_2 has two O atoms).

Step 2: grams H_2O_2 to moles H_2O_2.

$$50.0 \text{ g } \cancel{H_2O_2} \times \underbrace{\frac{1 \text{ mol } H_2O_2}{34.02 \text{ g } \cancel{H_2O_2}}}_{\text{molar mass of } H_2O_2} = 1.4699 \text{ mol } H_2O_2$$

Step 3: moles H_2O_2 to moles O_2. Have H_2O_2 (coefficient **2**), want O_2 (coefficient **1**).

$$1.4699 \text{ } \cancel{\text{mol } H_2O_2} \times \underbrace{\frac{1 \text{ mol } O_2}{2 \text{ } \cancel{\text{mol } H_2O_2}}}_{\text{coefficient of } O_2 \text{ / coefficient of } H_2O_2} = 0.7349 \text{ mol } O_2$$

Step 4: moles O_2 to grams O_2.

$$0.7349 \text{ } \cancel{\text{mol } O_2} \times \underbrace{\frac{32.00 \text{ g } O_2}{1 \text{ } \cancel{\text{mol } O_2}}}_{\text{molar mass of } O_2} = \boxed{23.5 \text{ g } O_2}$$

Why the other choices are wrong:

- a) 0.735 g: That is the **moles** of O_2. You stopped before the last step.
- b) 11.8 g: Used 16.00 g/mol (one O atom) instead of 32.00 g/mol for O_2.
- d) 47.0 g: Assumed 1 : 1. It takes 2 H_2O_2 to make 1 O_2.
- e) 50.0 g: Assumed the product mass equals the reactant mass. Some of the mass goes to H_2O.

6. Grams to moles of another substance — c) 0.998 mol

Source: Chapter 9 post, slide 18 (Stoichiometry Practice 2: moles of H_2SO_4 from 19.9 g KOH), slide 14
Homework match: HW22 — hw22ex3.png (moles of Br_2 needed to react with 0.206 g Al); HW23 — hw23ex1.png (CaC_2 + H_2O)
What it tests: Stopping at moles when the question asks for moles: g → mol → mol, no final mass step.

Step 1: molar mass of CaC_2. $40.08 + 2(12.01) = 64.10$ g/mol.

Step 2: grams CaC_2 to moles CaC_2.

$$32.0 \text{ g } \cancel{\text{CaC}_2} \times \underbrace{\frac{1 \text{ mol } \text{CaC}_2}{64.10 \text{ g } \cancel{\text{CaC}_2}}}_{\text{molar mass of } \text{CaC}_2} = 0.4992 \text{ mol } \text{CaC}_2$$

Step 3: moles $\mathbf{CaC_2}$ to moles $\mathbf{H_2O}$. Have CaC_2 (coefficient **1**), want H_2O (coefficient **2**). The question asks for **moles**, so stop here.

$$0.4992 \text{ } \cancel{\text{mol CaC}_2} \times \underbrace{\frac{2 \text{ mol } \text{H}_2\text{O}}{1 \text{ } \cancel{\text{mol CaC}_2}}}_{\text{coefficient of } \text{H}_2\text{O} \text{ / coefficient of } \text{CaC}_2} = \boxed{0.998 \text{ mol } \text{H}_2\text{O}}$$

Why the other choices are wrong:

- a) 0.250 mol: Ratio upside down: $\times$ 1/2 instead of $\times$ 2/1.
- b) 0.499 mol: Assumed 1 : 1. Water's coefficient is 2.
- d) 2.00 mol: Divided the molar mass by the grams (64.10 $\div$ 32.0).
- e) 18.0 mol: Multiplied by the molar mass of H_2O. That gives grams of water, not moles.

7. Balance first, then grams of a reactant — e) 16.0 g

Source: Chapter 9 post, slide 16 (why the equation must be balanced first), slide 9 (balancing a combustion equation), slide 19
Homework match: HW22 — hw22ex3.png (unbalanced equations such as Al + Br_2 $\rightarrow$ $AlBr_3$: balance, then convert)
What it tests: Balancing before using coefficients, then g $\rightarrow$ mol $\rightarrow$ mol $\rightarrow$ g for a second reactant.

Step 1: balance. P_4O_{10} has 4 P, so put **4** in front of P. It has 10 O, and O_2 supplies O in pairs, so put **5** in front of O_2 ($5 \times 2 = 10$). Balanced: $\mathbf{4P + 5O_2 \rightarrow P_4O_{10}}$.

Step 2: grams P to moles P. Atomic mass of P = 30.97 g/mol.

$$12.4 \text{ } \cancel{\text{g P}} \times \underbrace{\frac{1 \text{ mol P}}{30.97 \text{ } \cancel{\text{g P}}}}_{\text{molar mass of P}} = 0.4004 \text{ mol P}$$

Step 3: moles P to moles $\mathbf{O_2}$. Have P (coefficient **4**), want O_2 (coefficient **5**).

$$0.4004 \text{ } \cancel{\text{mol P}} \times \underbrace{\frac{5 \text{ mol } \text{O}_2}{4 \text{ } \cancel{\text{mol P}}}}_{\text{coefficient of } \text{O}_2 \text{ / coefficient of P}} = 0.5005 \text{ mol } \text{O}_2$$

Step 4: moles $\mathbf{O_2}$ to grams $\mathbf{O_2}$. Molar mass of O_2 = 2(16.00) = 32.00 g/mol.

$$0.5005 \text{ } \cancel{\text{mol O}_2} \times \underbrace{\frac{32.00 \text{ g } \text{O}_2}{1 \text{ } \cancel{\text{mol O}_2}}}_{\text{molar mass of } \text{O}_2} = \boxed{16.0 \text{ g } \text{O}_2}$$

Why the other choices are wrong:

- a) 0.500 g: That is the **moles** of O_2.

- b) 8.01 g: Used 16.00 g/mol (one O atom) instead of 32.00 g/mol for O_2.
- c) 10.2 g: Ratio upside down: $\times$ 4/5 instead of $\times$ 5/4.
- d) 12.8 g: Used the unbalanced equation (1 : 1). Balance it first.

8. Limiting reactant and leftover — a) CO is limiting, and 0.5 mol of O_2 is left over

Source: Chapter 9 post, slide 23 (Limiting Reactants Practice: 3Mg + N_2, both given in moles), slides 21–22
Homework match: HW23 — hw23ex2.png (determine which reactant is limiting)
What it tests: Comparing what one reactant needs with what is available, using the mole ratio, not raw moles.

Step 1: both amounts are already in moles. CO 3.0 mol, O_2 2.0 mol.

Step 2: how much O_2 does all the CO need? Have CO (coefficient **2**), want O_2 (coefficient **1**).

$$3.0 \ \cancel{\text{mol CO}} \times \underbrace{\frac{1 \text{ mol } O_2}{2 \ \cancel{\text{mol CO}}}}_{\text{coefficient of } O_2 \ / \ \text{coefficient of CO}} = 1.5 \text{ mol } O_2 \text{ needed}$$

Step 3: compare. You **have** 2.0 mol O_2 and only **need** 1.5 mol. O_2 is in excess, so **CO is limiting**.

Step 4: leftover. 2.0 mol O_2 available − 1.5 mol used = **0.5 mol O_2 left over**.

O_2 has fewer moles (2.0 vs 3.0) and is still the excess reactant. That's why you never compare raw moles.

Why the other choices are wrong:

- b) Neither is limiting; both are used up completely: That needs exactly a 2 : 1 mole ratio. 3.0 : 2.0 is not 2 : 1.
- c) O_2 is limiting, because there are fewer moles of O_2 than of CO: Compared raw moles. The equation needs only 1 O_2 per 2 CO, so 3.0 mol CO uses just 1.5 mol O_2.
- d) CO is limiting, and 1.0 mol of O_2 is left over: Right limiting reactant, wrong leftover. CO uses 1.5 mol O_2, so 2.0 − 1.5 = 0.5 mol remains.
- e) O_2 is limiting, and 1.0 mol of CO is left over: 2.0 mol O_2 would need 4.0 mol CO, and there is only 3.0 mol. CO runs out first.

9. Theoretical yield from two masses — b) 43.9 g

Source: Chapter 9 post, slide 27 (theoretical yield when both reactant masses are given), slide 24 (limiting reactant from grams), slide 25
Homework match: HW23 — hw23ex3.png (Fe + O_2 unbalanced: find the limiting reactant, then the mass of product)
What it tests: Balance → moles of each reactant → limiting reactant → product from the limiting reactant only.

Step 1: balance. $AlCl_3$ has 3 Cl and Cl_2 comes in pairs, so use 2$AlCl_3$ (6 Cl) and 3Cl_2 (6 Cl). Then 2 Al on the left: **$2Al + 3Cl_2 \rightarrow 2AlCl_3$**.

Step 2: grams to moles for each reactant. Al = 26.98 g/mol; Cl_2 = 2(35.45) = 70.90 g/mol.

$$10.0 \text{ g Al} \times \underbrace{\frac{1 \text{ mol Al}}{26.98 \text{ g Al}}}_{\text{molar mass of Al}} = 0.3706 \text{ mol Al}$$

$$35.0 \text{ g } Cl_2 \times \underbrace{\frac{1 \text{ mol } Cl_2}{70.90 \text{ g } Cl_2}}_{\text{molar mass of } Cl_2} = 0.4937 \text{ mol } Cl_2$$

Step 3: how much Cl_2 does all the Al need? Have Al (coefficient **2**), want Cl_2 (coefficient **3**).

$$0.3706 \text{ mol Al} \times \underbrace{\frac{3 \text{ mol } Cl_2}{2 \text{ mol Al}}}_{\text{coefficient of } Cl_2 \text{ / coefficient of Al}} = 0.5560 \text{ mol } Cl_2 \text{ needed}$$

Step 4: compare. You **need** 0.5560 mol Cl_2 but only **have** 0.4937 mol. The Cl_2 runs out first, so **Cl_2 is limiting** (even though Al has fewer moles and less mass).

Step 5: product from the limiting reactant. Have Cl_2 (coefficient **3**), want $AlCl_3$ (coefficient **2**). Molar mass of $AlCl_3$ = 26.98 + 3(35.45) = 133.33 g/mol.

$$0.4937 \text{ mol } Cl_2 \times \underbrace{\frac{2 \text{ mol } AlCl_3}{3 \text{ mol } Cl_2}}_{\text{coefficient of } AlCl_3 \text{ / coefficient of } Cl_2} \times \underbrace{\frac{133.33 \text{ g } AlCl_3}{1 \text{ mol } AlCl_3}}_{\text{molar mass of } AlCl_3} = \boxed{43.9 \text{ g } AlCl_3}$$

Cross-check: all the Al would make 0.3706 mol $AlCl_3$ × 133.33 = 49.4 g. The smaller amount (43.9 g, from Cl_2) is the theoretical yield.

Why the other choices are wrong:

- a) 0.329 g: That is the **moles** of $AlCl_3$.
- c) 45.0 g: Added the two reactant masses. Some Al is left over, so it isn't all in the product.
- d) 49.4 g: Calculated from Al. Al is in excess here, so this is more than the Cl_2 can make.
- e) 65.8 g: Used Cl_2 but assumed 1 : 1. The ratio is 2 $AlCl_3$: 3 Cl_2.

10. Percent yield — d) 88.5%

Source: Chapter 9 post, slide 30 (percent yield = actual / theoretical × 100%), slide 32 ($2KClO_3$ decomposition), slides 25, 28
Homework match: HW24 — hw24ex1.png (percent yield from theoretical and actual mass of magnesium oxide)
What it tests: Two parts: theoretical yield from the only reactant, then actual / theoretical × 100.

Part 1: theoretical yield of CaO. $CaCO_3$ is the only reactant, so it is limiting. Molar masses: $CaCO_3$ = 40.08 + 12.01 + 3(16.00) = 100.09 g/mol; CaO = 40.08 + 16.00 = 56.08 g/mol. Mole ratio: have $CaCO_3$ (coefficient **1**), want CaO (coefficient **1**).

$$25.0 \text{ g } CaCO_3 \times \underbrace{\frac{1 \text{ mol } CaCO_3}{100.09 \text{ g } CaCO_3}}_{\text{molar mass of } CaCO_3} = 0.2498 \text{ mol } CaCO_3$$

$$0.2498 \cancel{\text{ mol } CaCO_3} \times \underbrace{\frac{1 \text{ mol CaO}}{1 \cancel{\text{ mol } CaCO_3}}}_{\text{coefficients}} \times \underbrace{\frac{56.08 \text{ g CaO}}{1 \cancel{\text{ mol CaO}}}}_{\text{molar mass of CaO}} = 14.01 \text{ g CaO (theoretical)}$$

Part 2: percent yield. The actual yield (12.4 g, measured) goes on top; the theoretical yield on the bottom.

$$\text{percent yield} = \frac{12.4 \cancel{\text{ g CaO}} \text{ (actual)}}{14.01 \cancel{\text{ g CaO}} \text{ (theoretical)}} \times 100\% = \boxed{88.5\%}$$

Why the other choices are wrong:

- a) 0.885%: Forgot to multiply by 100.
- b) 49.6%: Divided by the mass of $CaCO_3$. The bottom must be the theoretical yield of **CaO**.
- c) 56.0%: That is the mass percent of CaO in $CaCO_3$ (56.08 / 100.09). It doesn't use the actual yield.
- e) 113%: Upside down: theoretical / actual. A yield can't be over 100% (slide 29).